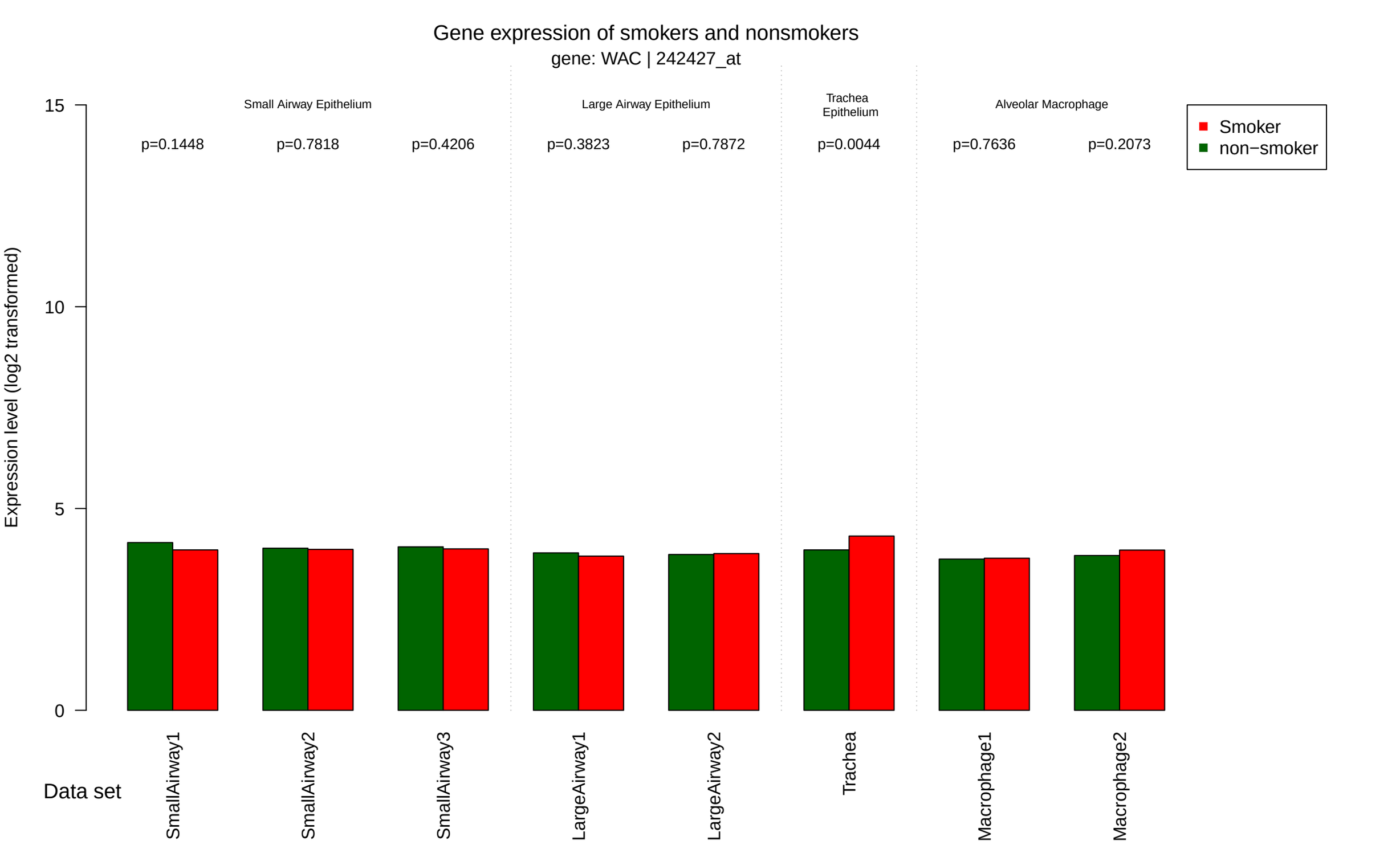
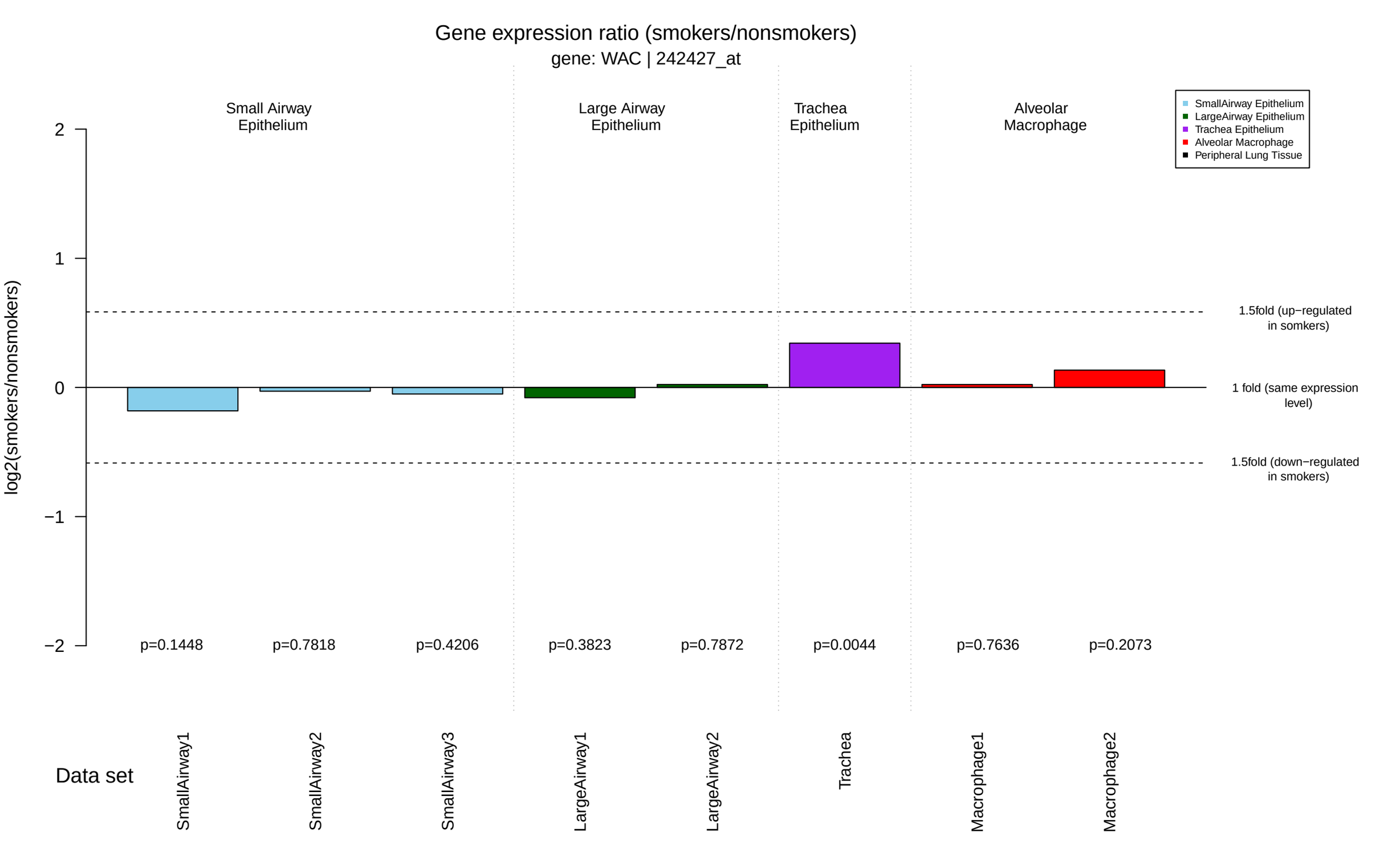
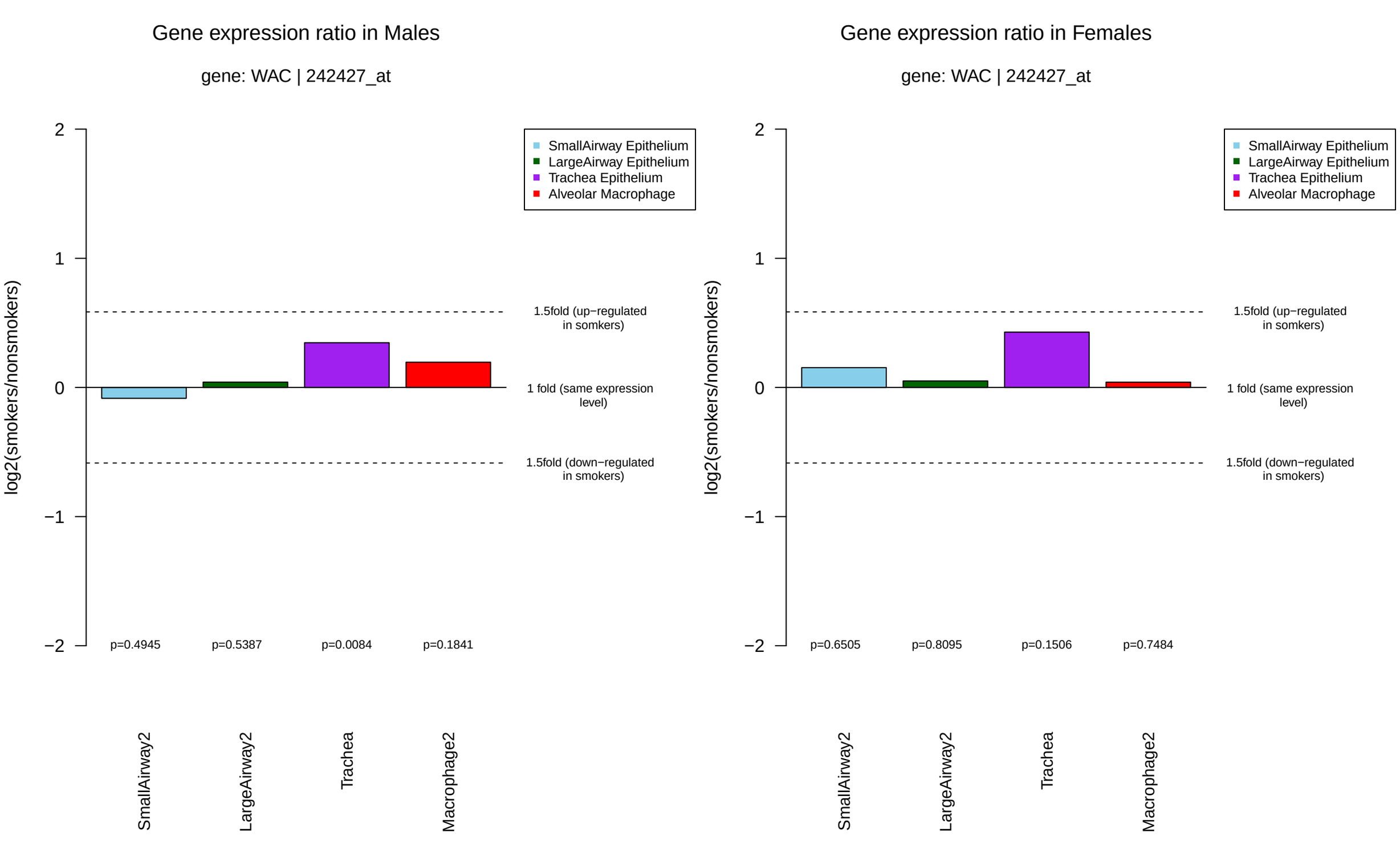
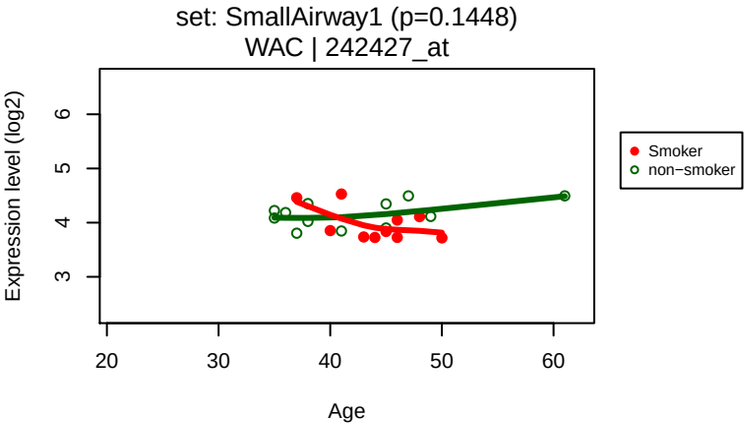
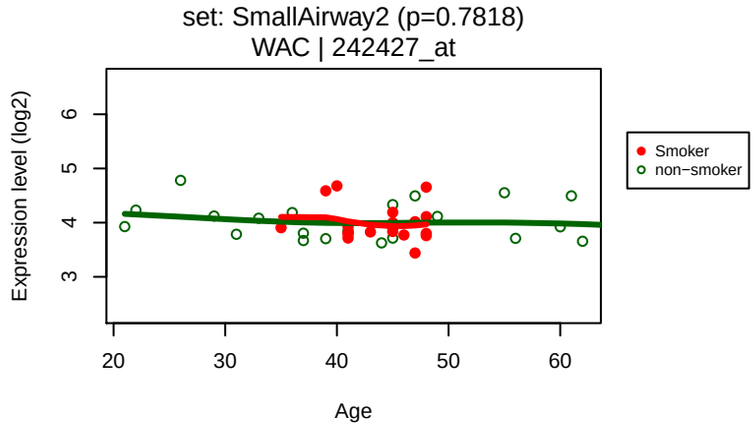
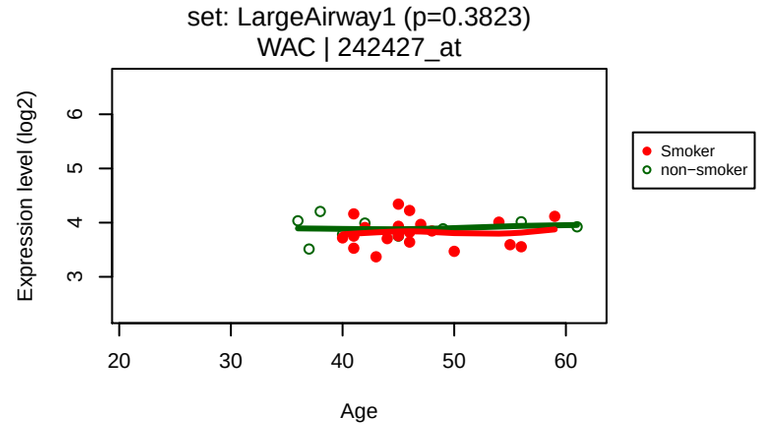

Gene expression of smokers and nonsmokers
gene: WAC | 242427_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1448
p=0.7818
p=0.4206
p=0.3823
p=0.7872
p=0.0044
p=0.7636
p=0.2073
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

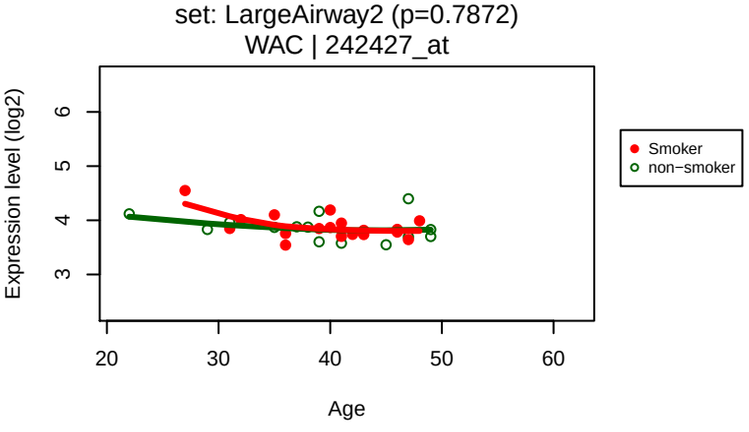

Gene expression ratio (smokers/nonsmokers)
gene: WAC | 242427_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1448
p=0.7818
p=0.4206
p=0.3823
p=0.7872
p=0.0044
p=0.7636
p=0.2073
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

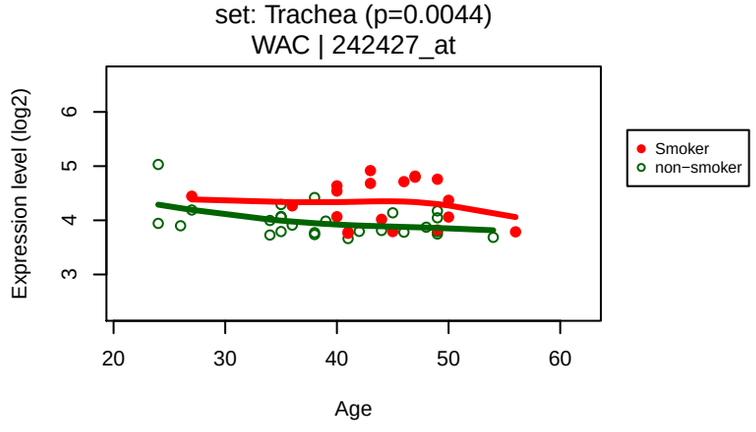

Gene expression ratio in Males
gene: WAC | 242427_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4945
p=0.5387
p=0.0084
p=0.1841
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WAC | 242427_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6505
p=0.8095
p=0.1506
p=0.7484
SmallAirway2
LargeAirway2
Trachea
Macrophage2

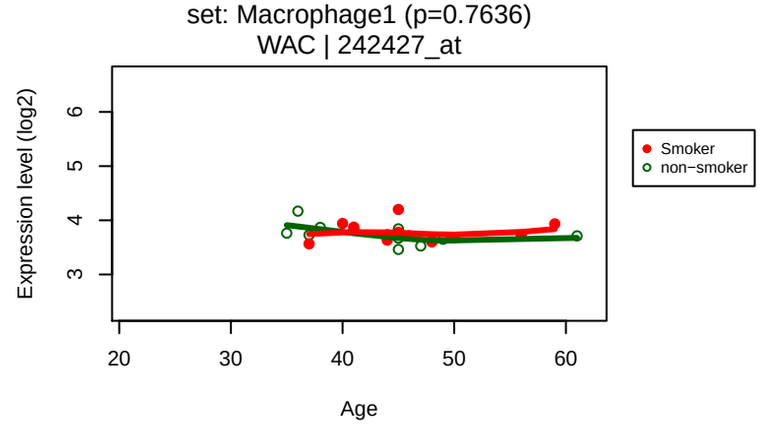

set: SmallAirway1 (p=0.1448)
WAC | 242427_at
Expression level (log2)
Age
Smoker
non−smoker

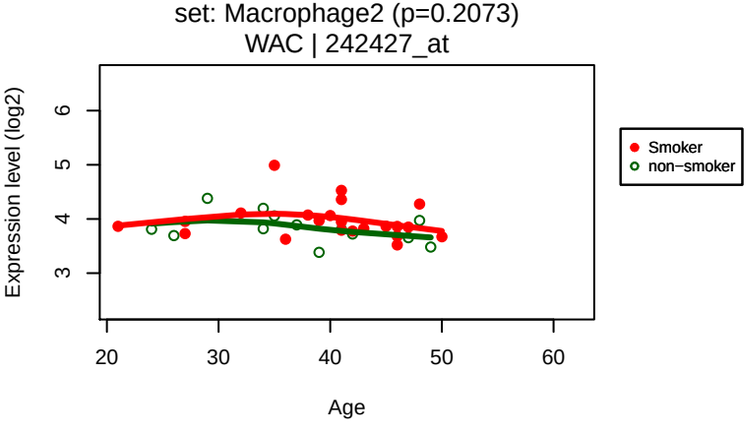

set: SmallAirway2 (p=0.7818)
WAC | 242427_at
Expression level (log2)
Age
Smoker
non−smoker

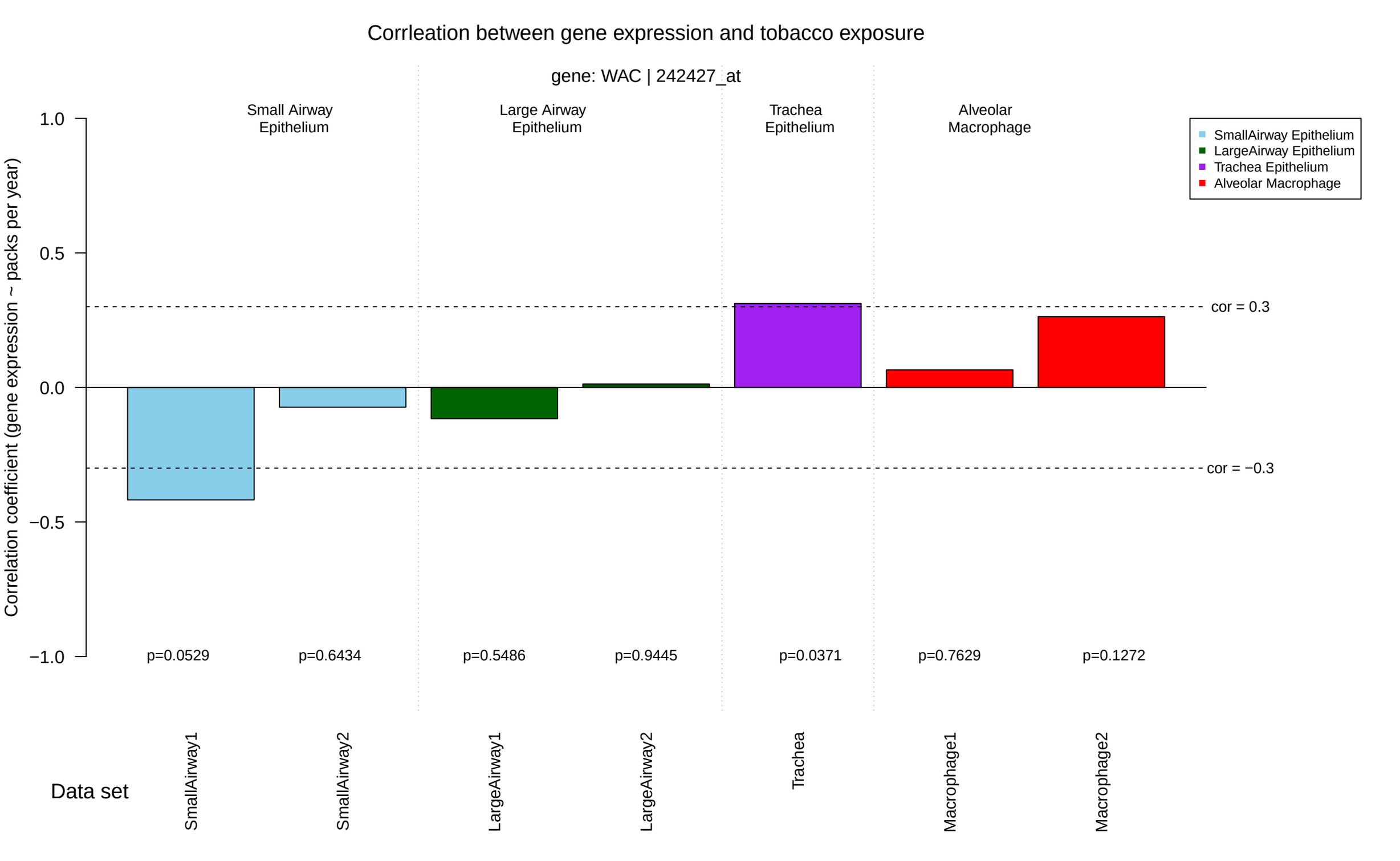

set: LargeAirway1 (p=0.3823)
WAC | 242427_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7872)
WAC | 242427_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0044)
WAC | 242427_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7636)
WAC | 242427_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2073)
WAC | 242427_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WAC | 242427_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0529
p=0.6434
p=0.5486
p=0.9445
p=0.0371
p=0.7629
p=0.1272
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2